AI-based Email Assistant

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Abstract

Nowadays, people receive numerous emails daily, and responding to all of them is a time-consuming task. To address this, we developed an AI-based system that helps in writing email replies automatically. Given an incoming email message, the system reads and generates a smart, professional reply. The backend uses Spring Boot and Spring AI to orchestrate model calls and business logic. Key techniques include email preprocessing, intent classification, clarifying-question generation, draft suggestion and refinement, and final reply generation via fine-tuned large language models. The system aims to save user time, reduce grammatical and tonal errors, and enable faster, more consistent communication across professional contexts. Evaluation will measure reply accuracy, time savings, and user satisfaction.

Keywords

Artificial Intelligence, Natural Language Processing, Email Automation, Large Language Models, Spring Boot, Productivity

Introduction

Background and Motivation: Email is one of the most popular modes of communication on the web. Replying to a large volume of emails often takes a significant amount of time. This motivated the creation of an AI assistant to help users reply faster while preserving a professional tone. Problem Statement: Writing emails manually is time-consuming and error-prone. Users often make grammatical mistakes or write unprofessional messages. The proposed system seeks to automate and improve the quality of email replies. Objectives: To develop a smart email assistant that can read incoming emails, understand intent and context, and generate replies automatically.

Literature Survey / Review

Summary of Related Work: Early research such as 'Intelligent Email' (Dredze et al., 2008) focused on assisting users with keyword identification, reply suggestions, and attachment reminders. More recent work (Miura et al., 2025) uses AI-generated questions to guide formal email composition with LLMs. Identified Gaps: Older systems used rule-based or shallow statistical methods and lacked deep contextual understanding. Modern LLM approaches improve naturalness but can reduce user control, require high-quality prompts, and depend heavily on the underlying model's reliability.

Literature Survey (continued)

Representative works: [1] Dredze, M., Wallach, H. M., Puller, D., Brooks, T., Carroll, J., Magarick, J., Blitzer, J., & Pereira, F. (2008). Intelligent Email: Aiding Users with AI. Proceedings of the Twenty-Third AAAI Conference on Artificial Intelligence, pp. 1524–1527. [2] Miura, Y., et al. (2025). Understanding and Supporting Formal Email Exchange by Answering AI-Generated Questions. arXiv:2502.03804. These works shaped our approach by highlighting the need for contextual models and user-in-the-loop refinement.

Problem Definition

Email overload and time-consuming replies create productivity bottlenecks. Users face unclear messages requiring back-and-forth exchanges and lack intelligent support in standard clients. This project targets automated analysis, summarization, prioritization, and drafting of context-aware replies to reduce cognitive load and improve turnaround time.

Objectives

1. Automate professional email reply drafting. 2. Maintain high linguistic quality and appropriate tone. 3. Provide clarifying-question generation when emails lack context. 4. Integrate with Gmail via a Chrome extension and secure REST APIs. 5. Enable continuous learning from user feedback.

Methodology / Proposed Work

Proposed Methodology: The approach involves a five-step pipeline: Email Preprocessing: Clean and normalize text (tokenization, stop-word removal, lemmatization) and extract metadata (sender, subject, thread history). Semantic Understanding: Use transformer models (BERT, RoBERTa) for intent classification and entity extraction to determine whether the email is a query, update, request, or complaint. Question Generation: When the email lacks key details, a transformer-based question generator (e.g., using Gemini/OpenAI) creates context-aware clarifying questions to reduce ambiguity. Draft Suggestion & Refinement: Summarization and style-adaptation modules produce an initial draft; grammar and tone-checkers refine it to professional standards. Reply Generation: Fine-tuned generative LLMs (GPT/Gemini) produce the final human-like reply. Reinforcement Learning from Human Feedback (RLHF) is used to continuously improve outputs based on user acceptance and edits.

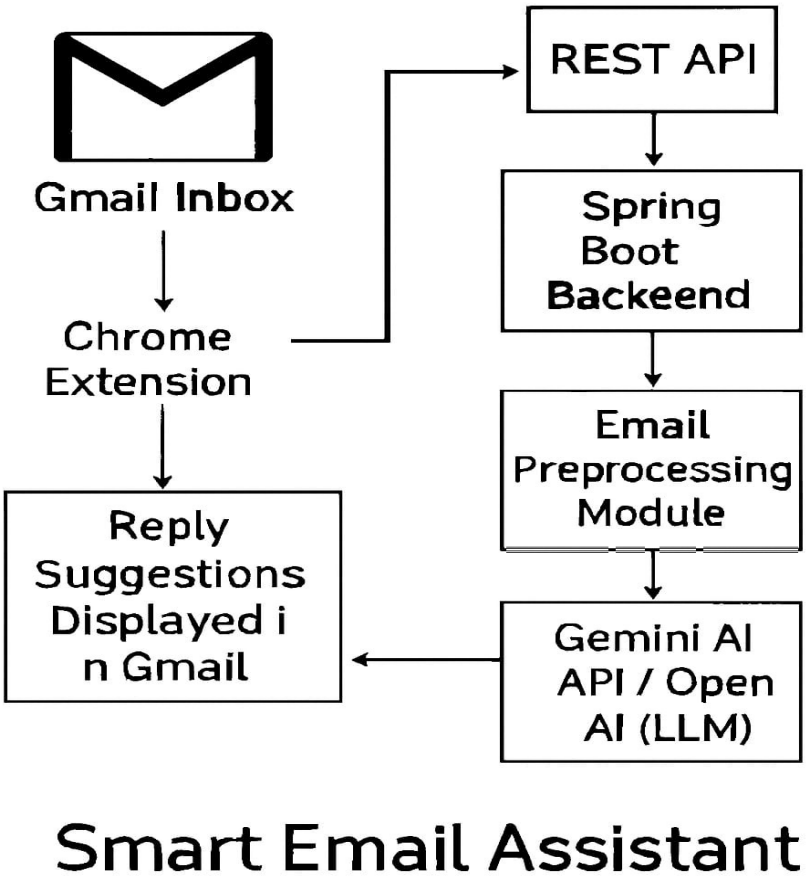

Figure: System Architecture - Smart Email Assistant

System Design / Architecture

The system uses a modular microservices architecture. Key components: - Email Integration Layer: Chrome extension connects to Gmail and forwards messages to REST endpoints (IMAP/SMTP alternatives possible for enterprise setups). - Preprocessing Module: Text cleaning, metadata extraction, thread summarization, and storage of context. - Inference Pipeline: Classification, summarization, question generation, and drafting services implemented as independent microservices. - Drafting Module: Applies style templates, tone control, and grammar correction. - Storage Layer: Secure database for user preferences, model checkpoints, and anonymized interaction logs.

Expected Outcome / Results

The assistant is expected to increase reply speed by ~3x, reduce time spent on email management, and maintain >85-90% accuracy in intent recognition. Evaluation metrics include BLEU/ROUGE for content overlap, human evaluation for tone appropriateness, and quantitative time-savings measured in user studies. Robustness tests will assess performance across varying email lengths, noise levels, and languages (where supported).

Applications

• Business and Corporate Use: Rapid handling of customer and internal communications. • Customer Support: Template-based responses adapted to context for consistent service. • Education: Faculty and administrative email management. • Personal Productivity: Assisting professionals and freelancers in managing correspondence. Scope: Focused on formal and functional emails; not designed for highly nuanced, sensitive, or legal communications without human oversight.

Ethical, Privacy, and Security Considerations

Privacy: Ensure emails are processed securely with encryption in transit and at rest, and provide opt-in consent for using cloud LLM APIs. Anonymization and retention policies will protect user data. Bias & Accuracy: Regular audits and human-in-the-loop review minimize biased or incorrect outputs. User Control: Final send-button remains with the user; the assistant suggests drafts rather than sending autonomously unless explicitly configured.

Conclusion

The AI-based Email Assistant offers a practical approach to reducing the time and effort involved in professional email communication. By combining preprocessing, semantic understanding, question generation, and fine-tuned LLM-based reply generation, the system balances automation with human oversight. The result is enhanced productivity, better communication quality, and a scalable architecture for future expansion (multilingual support, voice integration).

References

[1] M. Dredze et al., “Intelligent Email: Aiding Users with AI,” Proceedings of the Twenty-Third AAAI Conference on Artificial Intelligence, 2008, pp. 1524–1527. URL: https://www.aaai.org/Library/AAAI/aaai08contents.php [2] Y. Miura et al., “Understanding and Supporting Formal Email Exchange by Answering AI-Generated Questions,” arXiv preprint arXiv:2502.03804, 2025. DOI:10.48550/arXiv.2502.03804. URL: https://arxiv.org/abs/2502.03804 [3] ResearchGate link for Miura et al.: https://www.researchgate.net/publication/388791454_Understanding_and_Supporting_Formal_Email_Exchange_by_Answering_AI-Generated_Questions